

SOR3008: Census Income

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Introduction & Aim

Income level is a primary indicator of an individuals economic mobility and policy efficacy. The **\$50,000** threshold is a standard benchmark for middle class stability in the U.S., however income success rates (> \$50k) can differ significantly across demographics. This project utilises **data mining techniques** to predict whether an individuals income exceeds this threshold based on factors such as:

- **Demographics:** Age, Race, and Sex.
- **Socioeconomic Status:** Education level and Marital status.
- **Employment:** Occupation, Workclass, and Hours-per-week.

A key objective of this investigation is to identify the indicators of economic success through the evaluating the trade-off between **model interpretability** and **predictive power**. Isolating these high-impact features will aid to uncover the underlying structures that most reliably trigger entry into the high-income bracket.

Utilising the **UCI Adult Census Dataset**, the following predictive models are employed:

- **Naive Bayes** (Conditional independence assumption)
- **Logistic Regression** (Backward elimination).
- **Classification Trees** (CP pruning).
- **Random Forest Analysis** (Out-of-bag error and ensemble voting)
- **Neural Networks** (Multi-dimensional interactions)

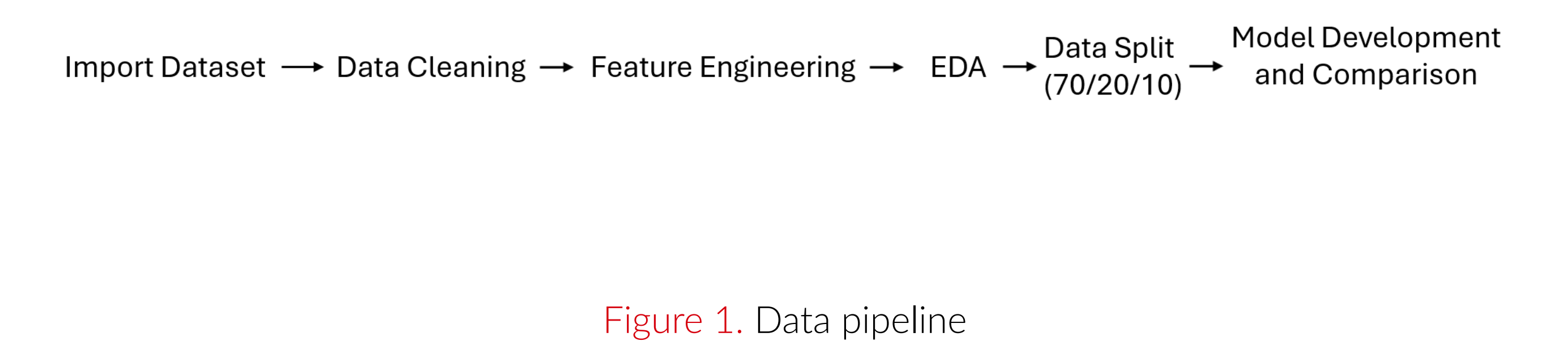


Figure 1. Data pipeline

Data & Methodology

This analysis uses the **Adult Census Income** dataset, sourced from the UCI Machine Learning Repository [?]. **SAS and R** code were employed to carry out data cleaning, exploratory analysis, and predictive modelling presented in this section.

The dataset consists of **32,561** observations across **14** socioeconomic attributes. The following table summarises the variable taxonomy by category:

Category	Variables	Description
Demographic	Age, Sex, Race	Personal characteristics
Employment	Workclass, Occupation, Hours-per-week	Professional sector and labor intensity
Social	Education-Num, Marital-Status, Relationship	Educational attainment and household structure
Geographic	Native-Country	Region of origin
Outcome	Income_Binary	Target: ≤50k (0) or >50k (1)

Figure 2. Variables in Census Dataset by Category

Missing Data

Initial diagnostics showed approximately **7%** of observations containing missing values, localised within the *Workclass* and *Occupation* attributes. These were addressed via **complete case analysis**, resulting in a final analytical sample of $N = 30,162$ observations.

- **Data Integrity:** Given the high cardinality of employment variables, imputation was avoided to prevent the artificial reduction of variance and potential bias in demographic correlations.
- **Statistical Power:** The remaining dataset is sufficiently large to ensure robust model performance and valid statistical inference without the need for synthetic data.

Feature Engineering & Categorisation

The primary objective of the data refinement phase was to **reduce model complexity** and prevent **overfitting**. By grouping rare or highly varying categories together, the amount data available for each group was increased, allowing for more reliable statistical estimates.

Variable	Original	Consolidated Groups	Primary Rationale
Occupation	14	White-Collar, Blue-Collar, Service, Low-Service, Protective	Military
Workclass	8	Private, Government, Self-Employed, Unemp.	Complexity Reduction
Country	41	North America, Europe, Asia, CentralAm, SouthAM, Other	Variance Control

Table 1. Strategic Consolidation of Categorical Predictors

Mathematical Feature Engineering:

- **Polynomial Age (Age^2):** Introduced in Logistic Regression to capture the non-linear "Inverse-U" career earnings curve.
- **Net_Capital:** Calculated as $Gain - Loss$. This reduces dimensionality and provides a unified signal for investment-based wealth.
- **Log Transformation:** Applied to skewed capital variables in PCA to normalize the distribution for variance analysis.

Model Validation Strategy

This project uses the **holdout method** [1, p. 219–222], an approach where the dataset is split into three mutually exclusive subsets:

- **Training Data:** 70% of the data is used to train the model. This teaches the model, allowing it to learn relationships and patterns
- **Validation Data (10%):** Used to fine-tune the model. This acts as a practice test to adjust settings and prevent overfitting during the training process.
- **Test Data:** 20% of the data is used to assess the models performance for a final evaluation

The holdout method was selected for its efficiency, speed and simplicity, particularly working with a very large dataset, with this split allowing a sufficient amount of data for training while still maintaining an adequate amount for consistent results.

Naive Bayes Methodology

A machine learning algorithm based on **Bayes theorem** holds the strong assumption that all features are independent of each other given the class.

$$P(C \mid X) = \frac{P(X \mid C) P(C)}{P(X)} \quad (1)$$

Logistic Regression Methodology

The baseline model estimates the relationship between categorical predictors and the **log-odds** of the success rate (> \$50k) using the logit link function:

$$\log \left(\frac{P(Y = 1)}{1 - P(Y = 1)} \right) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k \quad (2)$$

- **Beta (β):** Represents the change in log-odds for a one-unit change in the predictor.
- **Odds Ratio (e^β):** Quantifies the strength of association relative to the reference category. An $OR > 1$ indicates an increased likelihood of earning > \$50k.

Classification Tree

The tree is built in a greedy manner [2, p. 307] by selecting splits that maximize **Information Gain**—the reduction in **Entropy** (uncertainty). The formula for entropy [2, pp. 310–312] is:

$$\text{Entropy} = -p_+ \log_2(p_+) - p_- \log_2(p_-)$$

where p_+ and p_- are the proportions of solved and unsolved cases. The process recurses until stopping criteria are met. Variable significance is evaluated via:

- **Relative Importance:** The proportion of total entropy reduction attributed to a specific variable (ranging from 0 to 1).
- **Split Count:** The frequency with which a variable is used to split the data.

Random Forest Methodology

A Random Forest [4] is an ensemble learning method that constructs a forest of uncorrelated decision trees to improve predictive accuracy and control over-fitting through two key processes:

- **Bagging:** Each tree is trained on a random bootstrap sample of the data, reducing model variance.
- **Feature Subsampling:** At each split, only a random subset of predictors is considered, ensuring tree diversity.

The **final model output** is determined by the majority vote of all individual trees:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \dots, T_B(x)\} \quad (3)$$

Neural Network Methodology

A **Multi-Layer Perceptron architecture** captures complex non-linear interactions through interconnected nodes. Each node computes a **weighted sum** passed through an activation function (f):

$$z_i = f \left(\sum w_{ij} x_j + b_i \right) \quad (4)$$

- **Weights (w):** Adjusted via backpropagation to minimize the error function.
- **Activation Function:** Typically Hyperbolic Tangent or Logistic, allowing the model to learn multi-dimensional patterns.

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Exploratory Data Analysis

The goal of this analysis was to visualize how **demographic, occupational, and educational factors** relate to the likelihood of earning over \$50K. These insights validate the inclusion of specific predictors in our Logistic Regression and Neural Network models.

Category Group		N (Sample Size)	Proportion >50K (%)
Sex	Female	9,782	11.4%
	Male	20,380	31.4%
Race	Amer-Indian-Eskimo	286	11.9%
	Asian-Pac-Islander	895	27.7%
	Black	2,817	13.0%
	Other	231	9.1%
	White	25,933	26.4%

Table 2. Baseline demographic breakdown showing significant income disparities across Sex and Race ($N = 30,162$).

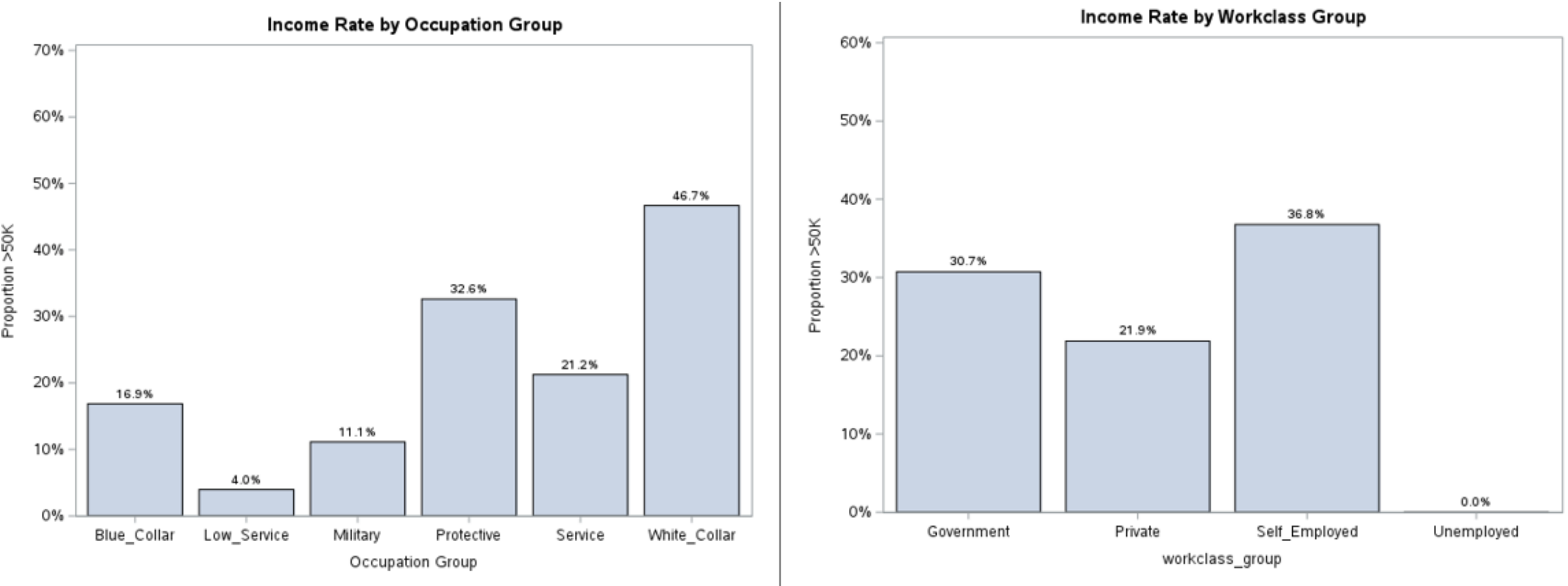


Figure 3. Economic Drivers: Income probability peaks in professional and managerial roles, while private sector and self-incorporated workclasses show the highest rates of high-income earners.

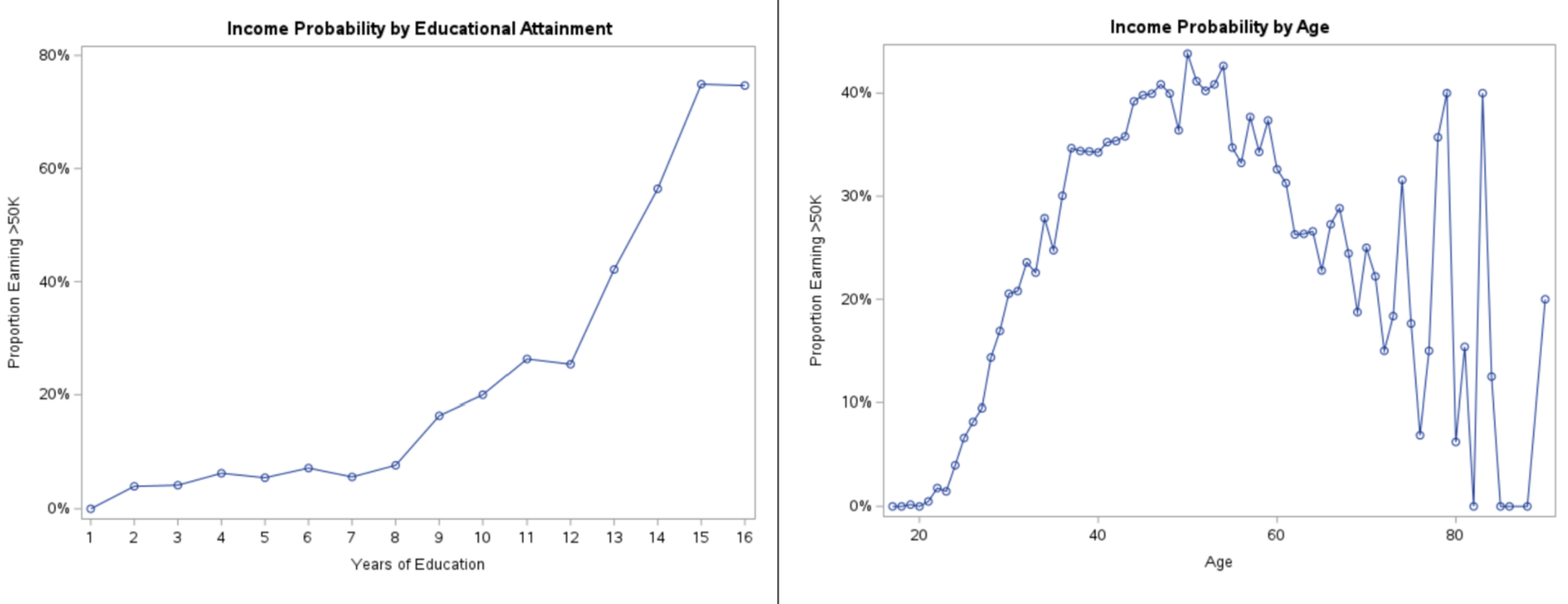


Figure 4. Human Capital Trends: Probability follows a non-linear quadratic relationship with Age (peaking at ≈ 50) and a discrete "step-wise" increase with years of formal education.

Pearson's Chi-Square tests were conducted for all categorical predictors (Sex, Race, Occupation, Workclass). All yielded $p < 0.0001$, confirming that the observed differences in income proportions are **statistically significant** and highly unlikely to be due to chance.

Principle Component Analysis

The structure of the four continuous numerical variables was examined: **age**, **education_num**, **net_capital**, and **hours_per_week**.

Before performing PCA, a **Log transformation** was applied to **net_capital**, due to its significant right-skewness. the formula $\text{sign}(x) \cdot \log(1 + |x|)$ was applied, stabilising the variance and preventing extreme outliers. Following this, **scaling** was applied, with all variables centred to have a **mean of zero and unit variance**, ensuring variables with larger scales did not overwhelm smaller-ranged variables.

Principle components of importance:

- **PC1:** Representing an aggregate of **labour and education**, this component yielded strong negative loadings across all variables, suggesting that an increase in these inputs caused a decrease in PC1 score.
- **PC2:** The strong driver from this component was **net_capital**. This variable isolates wealth accumulation as a distinct source from other factors such as education or hours worked.

This classifier was unable to include the categorical variables of the dataset, giving it a clear limitation. This is clear in the models discrepancy between its high **Specificity (0.9297)** and its more moderate **Sensitivity (0.6091)**.

Evaluation Theory

Model efficacy is assessed using metrics derived from the confusion matrix and Receiver Operating Characteristic (ROC) analysis.

- **Accuracy, Sensitivity, and Specificity [2, p. 307]:** Evaluation of classification success at a fixed 0.50 threshold.
$$\text{Sens} = \frac{TP}{TP + FN} \quad ; \quad \text{Spec} = \frac{TN}{TN + FP}$$
- **Area Under the Curve (AUC [2, p. 147]):** Measures the model's ability to discriminate between classes. Scores above **0.80** are considered excellent.

* TP/TN : True Pos/Neg; FP/FN : False Pos/Neg; AUC: Area Under ROC.

Logistic Regression & Model Selection

Logistic regression [1] is used to estimate the probability of the binary income outcome ($> \$50K$ vs. $\leq \$50K$). The model was refined via **Backward Elimination** based on the **Wald Chi-Square** statistic to ensure only significant predictors ($p < 0.05$) were retained. The fit was further improved by including an Age^2 term to capture the non-linear earning trajectory observed during EDA, resulting in a Final **SC (BIC)** of 15,379.2.

Category	Comparison (vs. Ref)	β	Odds Ratio	p-value
Education	Continuous (Years)	0.2912	1.338	<.0001
Occupation (Ref: Service)	White-Collar vs. Service	0.4262	1.531	<.0001
	Blue-Collar vs. Service	-0.6395	0.528	<.0001
	Low-Service vs. Service	-1.2406	0.289	<.0001
Marital Stat. (Ref: Never-married)	Married-Civ vs. Never	2.3380	10.361	<.0001
	Divorced vs. Never	0.2311	1.260	0.0114
	Widowed vs. Never	0.5219	1.685	0.0025
Workclass (Ref: Private)	Self-Employed vs. Private	-0.1816	0.834	0.0022
	Government vs. Private	-0.2292	0.795	0.0001
Sex	Male vs. Female	0.3162	1.372	<.0001

Table 3. Logistic Regression Coefficients and Odds Ratios (e^β)

On the test set ($n = 6,033$), the model achieved **82.5% Accuracy** and an **AUC of 0.886**. While highly reliable at identifying low earners (**92.3% Specificity**), the model demonstrated moderate sensitivity to high earners (**52.7% Sensitivity**).

Classification Tree [2]

The initial tree model creating using rpart generated a tree with a **total of 59 nodes**, with 30 leaf nodes. the pruning of the tree was carried out using the **1-SE rule**, which found the minimal ex-error to be **0.5598258**. This selects the smallest tree whose cross-validation error is within 1 SE of the minimum error, which was found to be **9 total nodes remaining**.

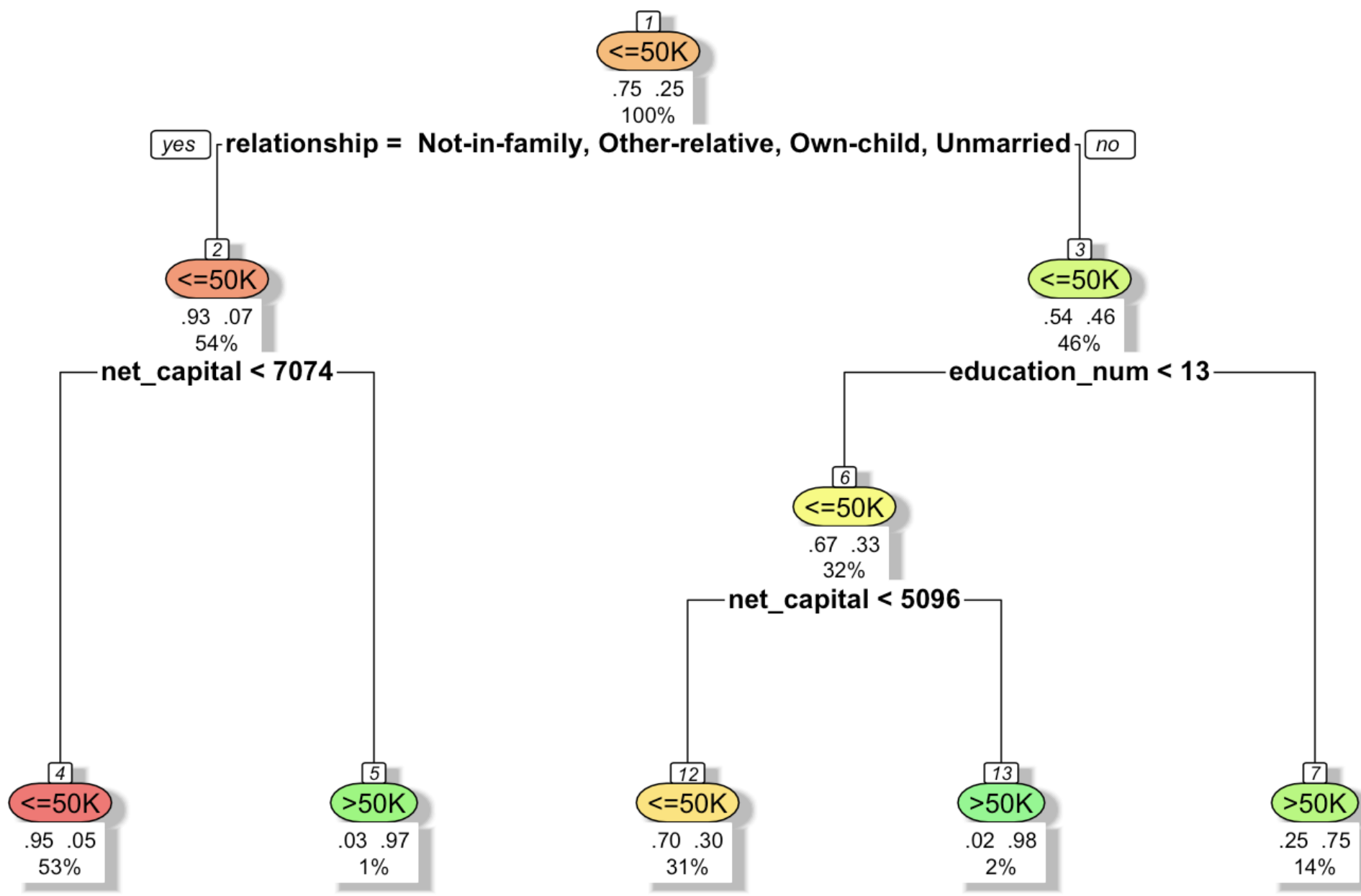


Figure 5. Pruned Classification Tree

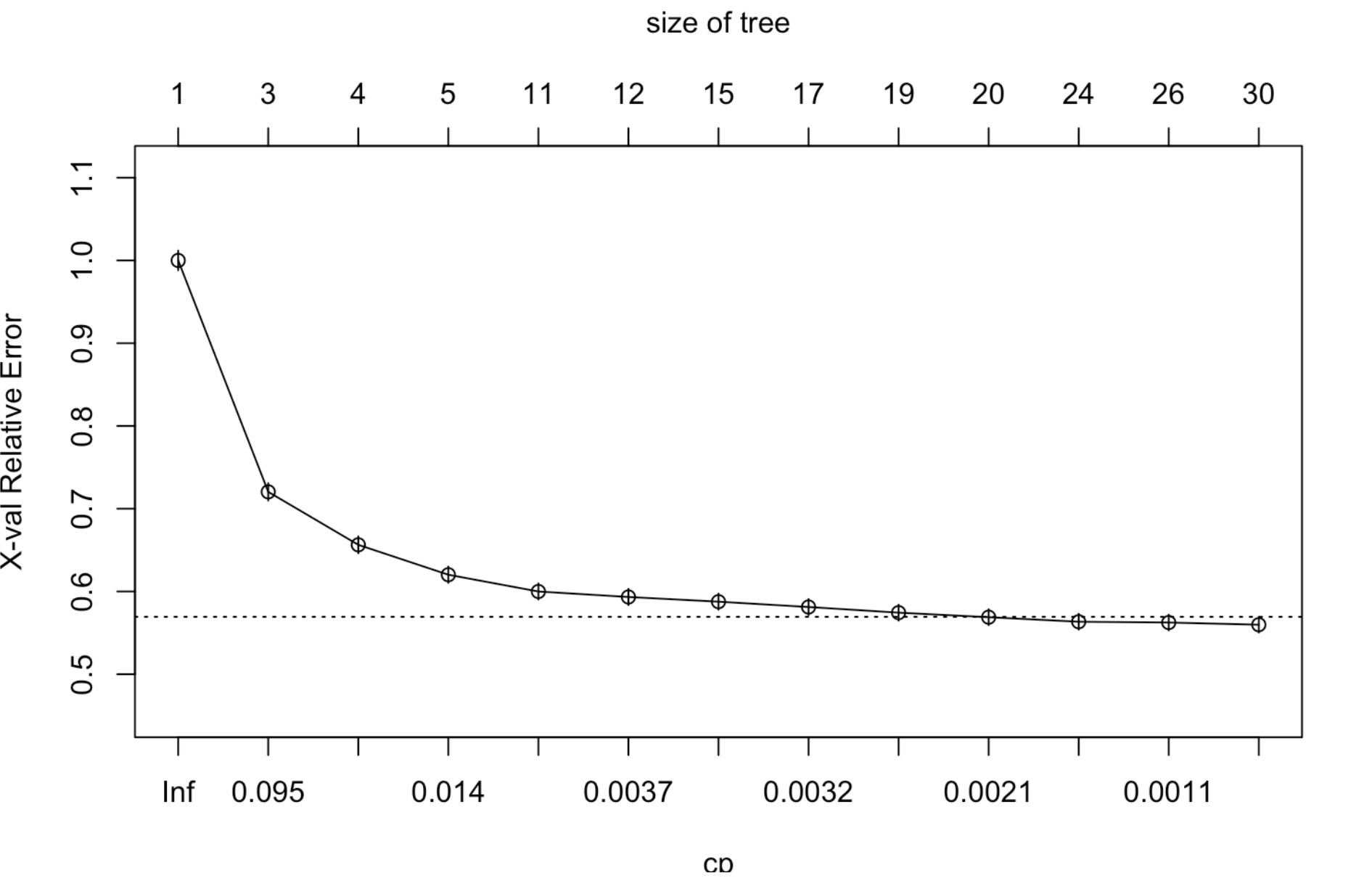


Figure 6. CP plot, dotted line is the 1-SE rule

The importance table suggests that **relationship** and **marital status** are the most important predictors, with scores over 2000. the variable **country group** and **race** seem to have the **lowest importance** as predictors, with negligible scores, suggesting that these variable provide little unique information gain.

Table 4. Variable Importance Scores from Pruned Classification Tree

Variable	Importance Score
relationship	2258.65
marital_status	2225.09
net_capital	924.69
sex	760.06
education_num	739.50
occupation_group	564.23
age	530.47
hours_per_week	302.96
country_group	15.80
race	6.52

Random Forest Analysis

The **mean decrease accuracy** table demonstrates the loss of accuracy if the variables were randomly shuffled. The results from this table suggest that net capital is the most critical signal for the model, as it holds the highest drop.

The **mean decrease gini** plot shows the contribution of purity that a variable adds to all nodes across the tree. This plot shows net capital to once again be the highest, followed more closely by age and relationship. The **mean decrease accuracy** table shows education and occupation follow in value, suggesting that while age didn't hold as much accuracy as education, it is stronger in consistently splitting data into subgroups.

Random Forest Variable Importance

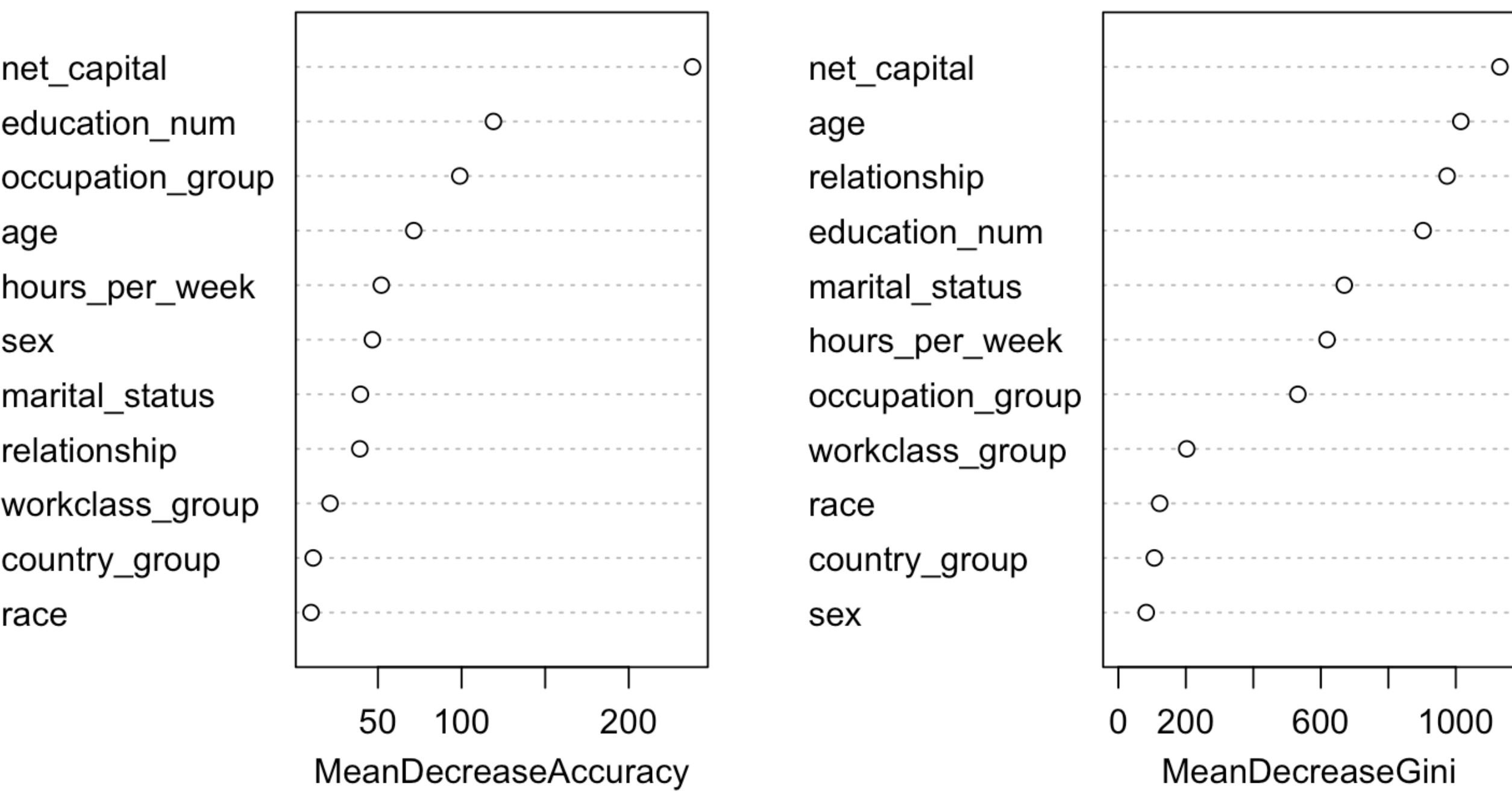


Figure 7. Pruned Classification Tree

Naive Bayes

The model produced highly polarized posterior probabilities. For the majority of the test observations, the class probabilities tended toward the **extremes of 0 or 1**. This is a characteristic result of the **"naive" assumption**; as multiple independent pieces of evidence (features) are multiplied, the resulting product often drives the probability to near-certainty.

However, **borderline cases** yielded a more moderate probability of, for example $P(\leq 50K) = 0.778$, indicating a region of the feature space where the attribute values provided conflicting evidence for the two income classes.

The Naive Bayes model achieved a **Test Accuracy of 79.6%**, demonstrating a strong alignment with the **Validation Accuracy of 79.7%**. This minimal generalization error suggests that the model is robust and has not over-fitted the training data.

Despite its simplicity, the model effectively captured the primary statistical signals within the dataset. The result suggests that the underlying distribution of features provides a solid probabilistic foundation for classification, even without accounting for complex inter-variable dependencies.

Neural Network

The final predictive modelling technique employed was a **neural network** [3]. Figure 8 shows a very simple model, only using the **numeric variables**. This is coded in R using the **neuralnet** function.

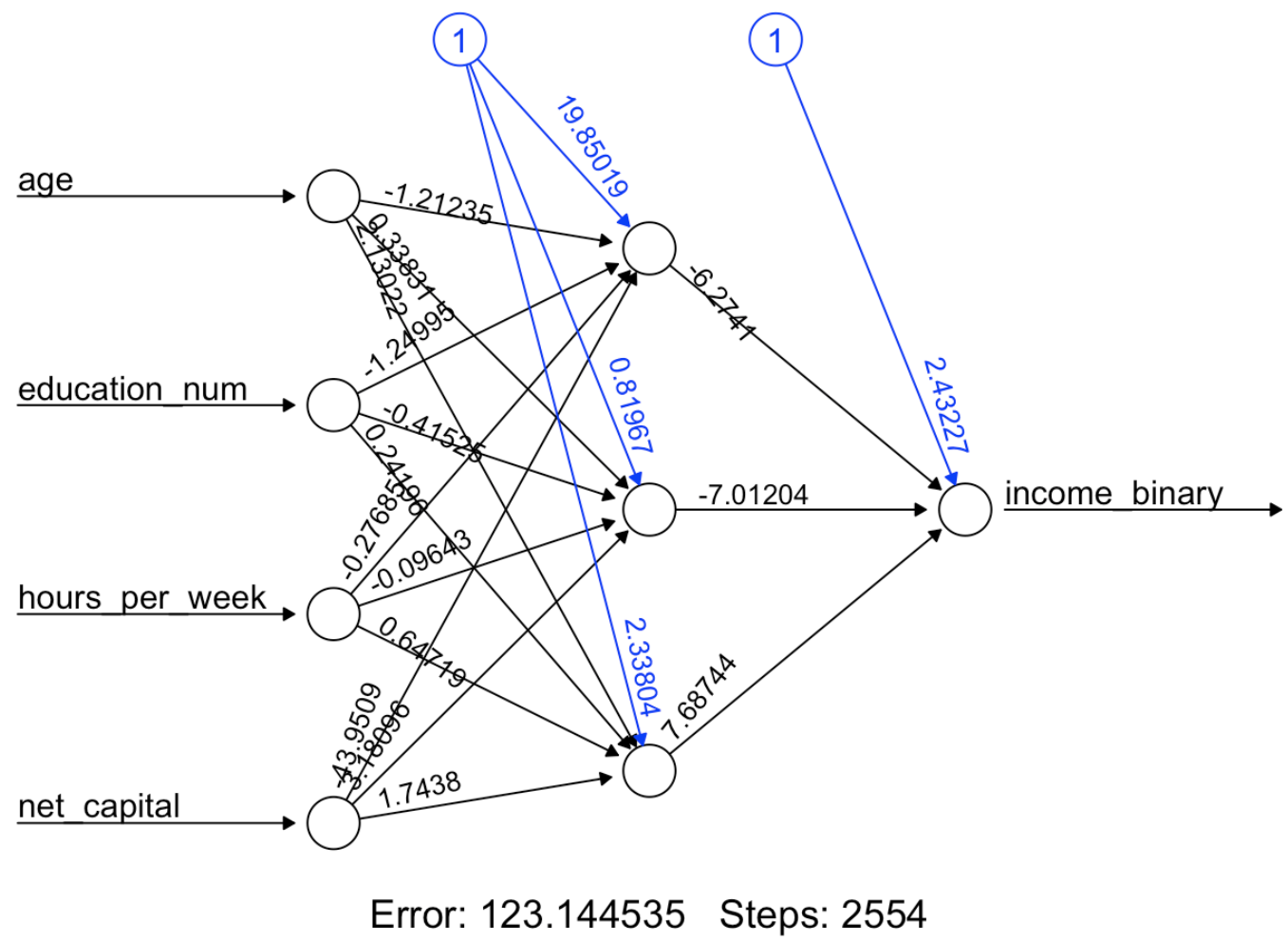


Figure 8. Visualisation of the process of the neural network, with the nodes from left to right demonstrating the input, hidden and output layers.

A more complex neural network, featuring all variables was coded on **SAS**, using the **hpneural** function. The confusion matrix below illustrates the **model's performance** on the unseen test data:

Actual	Predicted	
	≤\$50K (0)	> \$50K (1)
≤\$50K (0)	4,209	330
> \$50K (1)	636	858

Table 5. Confusion matrix for neural network model on test data (n=6,033)

Based on these results, the model achieved an overall **accuracy of 83.99%**, with a **sensitivity of 57.43%**, a **specificity of 92.73%**, and an **AUC of 0.901**. This indicates the model is highly effective at identifying income patterns, particularly for individuals earning ≤\$50K.

Comparison

The table below shows the performance metrics across all the different predictive models:

Model	Accuracy	Sensitivity	Specificity	AUC
Random Forest	0.850	0.608	0.929	0.892
Neural Network	0.840	0.574	0.927	0.901
Decision Tree	0.835	0.491	0.949	0.845
Logistic Regression	0.825	0.527	0.923	0.886
Naive Bayes	0.796	0.303	0.958	0.864
Null Model (Baseline)	0.751	0.000	1.000	0.500

Table 6. Consolidated Model Performance Comparison (Test Data)

This table demonstrates a **strong difference** in performance across the different models. This difference comes mainly from their ability to identify higher earners, as all models were **highly reliable** at identifying the majority class.

- All models achieved **Specificity >92%**. This means that the models rarely misclassified a low earner as a high earner.
- The **Neural Network** achieved the **highest AUC (0.901)**, indicating superior performance in capturing complex, non-linear interactions between variables.
- Naive Bayes** was the **weakest classifier**. This is because high correlations between variables are missed due to the assumption of independence.
- Random Forest** provided the highest overall **Accuracy (85.0%)**, making it the most balanced model for raw prediction.

Conclusion

In conclusion, this report has identified indicators of economic success using various machine learning classifier tools, aiming to balance interpretability and power. The results of these models showed random forest to provide the best predictive power with an accuracy of 85.0% and an AUC of 0.892. The key drivers were found to be Net Capital ,relationship status and education . these factors were found to be the strongest predictors across multiple models. while neural network and random forest were more accurate, logistic regression and the decision tree provided unique insights, such as specific odds ratios for different occupations.

Limitations

- Class Imbalance:** High Specificity (92%) vs. moderate Sensitivity (57%) confirms a model bias toward the majority class (≤\$50k earners).
- Independence Assumption:** The Naive Bayes model assumes all features are independent; in reality, variables like *Education* and *Occupation* are highly correlated.
- Temporal Decay:** Reliance on 1994 data assumes constant economic relationships; it cannot account for 30 years of inflation or labor market shifts.
- Self-Reporting Bias:** Census data is subject to inaccurate income figures and "prestige bias" in respondent-provided occupation/education levels.
- Complete Case Analysis:** Deleting missing observations (7%) assumes data is Missing Completely at Random (MCAR). If values are Missing at Random (MAR), this may introduce selection bias.

Further Work

- Granular Feature Analysis:** Re-evaluate models using the original categories for occupation, workclass and native-country and compare the results.
- Threshold Optimisation:** Compare performance across multiple probability thresholds (0.3–0.7) to identify optimal balances for specific utility [2].
- Advanced Validation:** Transition to k-fold cross-validation or bootstrap resampling to provide more robust error estimates than a single holdout split [1].
- Alternative Techniques:** Explore distance-based classification via **k-Nearest Neighbors** (k-NN) or clustering to identify case subgroups [5].
- Feature Expansion:** Test interaction terms [2] between key predictors (e.g., *Education* × *Occupation*).

References

[1] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.

[2] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning with Applications in R*. Springer.

[3] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.

[4] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>

[5] Altman, N. S. (1992). An introduction to kernel and nearest-neighbor nonparametric regression. *The American Statistician*, 46(3), 175–185. <https://doi.org/10.1080/00031305.1992.10475879>